Tutorial 9

The core exercises are considered part of the course material, and you are advised to at least finish these. For additional practice and interesting applications, advanced exercises are available as well – feel free to pick and choose from these.

Unless otherwise specified, we expect you to create your own algorithms for the exercises in each tutorial. This means that you should **not use libraries (e.g. `SciPy`, `numpy`)** for the methods that are part of the course material.

You are free to write and execute code on your own laptop, but please be aware that all code you hand in needs to run on the vdesk machines[1] out of the box (e.g. no installing additional packages), so you may want to get used to this already.

Core exercises

1. **Detecting Outliers**
   In this simple exercise we are going to consider five datasets which we will model using a linear relation.
   (a) Download the five datasets with `wget` from:

   http://home.strw.leidenuniv.nl/~stefanon/NUR_2021/T9_data/outliers_dataset<i>.txt
   where $< i >= 1..5$, and inspect their content with a plot.
   (b) Pick one of the dataset to your liking, fit a linear relation using least squares and compare the result to the data set.
   (c) As it can be seen , the fit does not properly reproduce the main trend of the data. Implement a robust estimate to minimize the impact of the outliers on the best fit parameters. Plot the result.
   (d) *(Optional)* Repeat the same analysis for the other four datasets.

2. **Kolmogorov-Smirnov test**
   In this exercise we are going to explore how the Kolmogorov-Smirnov test works. To setup, download the set of 12 random numbers from

   http://home.strw.leidenuniv.nl/~stefanon/NUR_2021/T9_data/RandomNumbers<i>.txt, where <i> indicates the number of the dataset.
   (a) Generate a Kolmogorov-Smirnov (KS) test function.
   (b) Write a function to search which of the random number dataset/s is consistent with a Cauchy distribution.
   (c) Repeat your exercise but only use the first 100 numbers in each dataset.
   (d) Write a function to determine using the KS test if there are pairs of datasets with very similar distributions. Do this with all the numbers you have in each set, and with 100 random numbers.
   (e) *(Optional)* Instead of doing the KS test, it is almost always better to generalize the KS test to include the absolute size of the most positive and the most negative difference between the two cumulative distribution functions. This is called the Kuiper's test. Apply the Kuiper's test to compare the distributions you downloaded[2]. Can we rule out more distributions that are not equivalent to each other?

3. **The chi-square test**
   In this problem, we will use Pearson's chi-square χ^2 statistics to test the goodness of a fit. The chi-square statistic can be expressed in the following form for discrete data points:

$$\chi^2 = \sum_{i=1}^{\nu} \frac{(x_i - \mu)^2}{\sigma_i^2} \tag{1}$$

[1] `https://helpdesk.strw.leidenuniv.nl/wiki/doku.php?id=manuals:virtualdesktopserver`
[2] For more information on the Kuiper's test see Chapter 14.3.4 in Numerical Recipes

where μ is the mean and σ_i^2 is variance.

(a) Five different groups at CERN have measured the mass of a new hypothetical particle. Download their measurements using `wget` from:

http://home.strw.leidenuniv.nl/~stefanon/NUR_2021/T9_data/Chi-square.txt.

The mass measurement of each group is given in no particular order. Implement a chi-square test function, and apply it to the data. Consider the reduced chi-squared statistic. What does the value imply here?

Hint:

$$\mu = \frac{\sum_i m_i/\sigma_i^2}{\sum_i 1/\sigma_i^2}, \tag{2}$$

$$\sigma_\mu^2 = \frac{1}{\sum_i 1/\sigma_i^2}. \tag{3}$$

(b) To check the significance of the goodness of fit (we fit with a constant value) in the previous part, calculate the P-value, i.e. the probability of finding a value higher than the χ^2 obtained above, given that the probability density function of the χ^2-distribution has the form:

$$f(\chi^2) = \frac{1}{2^{\nu/2}\Gamma(\nu/2)} e^{-\chi^2/2}(\chi^2)^{(-1+\nu/2)}. \tag{4}$$

Use the improper integration routine you have created in Tutorial 4 to estimate the value of the integral.

(c) *(Optional)* Download a new dataset from:

http://home.strw.leidenuniv.nl/~stefanon/NUR_2021/T9_data/Chi-square2.txt

and check the goodness of fit assuming a linear model $y = ax + b$ with $a = 0.16$ and $b = 0.83$. Compute the reduced χ^2 statistic and the P-value. What do these values tell us?

(d) *(Optional)* Perform a G-test for the above 2 problems. The test statistic is a chi-squared statistic with the same degrees-of-freedom as the analogous chi-squared test. You can use the same distribution table. What do the statistic and P-value signify here in comparison to the previously obtained chi-squared statistic?

Advanced exercises

4. **Galaxy stellar mass function – continued**

In last week's tutorial, we constructed the stellar mass function for galaxy clusters and fitted the data using a Schechter function. It is common practice to fit data with errors to improve the reliability of fitted parameters. In this problem, you will use bootstrapping to obtain multiple datasets from one given dataset and improve fitting by using the errorbars. Besides this we will also calculate the confidence interval using a grid.

For this exercise, use the same dataset as you have used in tutorial 8. It can be downloaded using `wget` from:
http://home.strw.leidenuniv.nl/~garcia/NUR/smfdata.txt.

(a) As you did last week, construct the stellar mass function (SMF) for this data, which is the number of galaxies per mass bin per cluster. Use a convenient number of bins for $\log_{10} M_*$ range [8.5, 12].

(b) Fit the stellar mass function using the Schechter function and plot the resulting curve and the original data. Adopt the following form of the Schechter function:

$$\Phi\left(\mathrm{cluster}^{-1}\mathrm{dex}^{-1}\right) = \ln(10)\cdot\Phi^*\cdot\left[10^{(M-M^*)(1+\alpha)}\right]\exp\left[-10^{(M-M^*)}\right], \tag{5}$$

where M^* is the characteristic mass, α is the low-mass slope, and Φ^* is the overall normalisation.

(c) Produce 50 realizations of the data-array containing the stellar mass, each time sampling N number of galaxies from the data with replacement, with N the total number of galaxies in the array. Make the SMF for each realization of data using the same mass bin range. Estimate the standard deviation for number of galaxies in each mass bin. Now fit the same data-array as before but also using the error values. Plot the SMF with errorbars and the fitted line.

(d) Now, produce another 50 realizations of the data array as before. But instead of sampling N number of galaxies each time, sample M number of galaxies, where M follows a Poisson distribution with mean equal to N. Do the rest of the steps as the previous question.

(e) Plot the data with the fitted lines from the above 3 procedures. Which one is better fitted by eye-estimation? Obtain reduced chi-squared value for each of them. Which fitting gives a more reliable statistic value?

(f) Now we are going to calculate the confidence interval on the galaxy stellar mass function. Select your best estimate values for Φ^*, M^* and α. Use your best fit using your eye-estimation.

(g) Generate two grids for your parameters which calculate the χ^2 of your model. The first grid should start at 75% of your parameters and extend until 125% of your parameters, the other should be extending from 50% of your parameters until 150% of your parameters. Use 100 points in each dimension, such that you create 2 grids of 100^3.

(h) Subtract your central value from your χ^2 grid to obtain a $\Delta\chi^2$ grid.

(i) For both grids create three plots of the $\Delta\chi^2$ values.

(j) If necessary improve your grid such that we have good range of $\Delta\chi^2$.

(k) Create the confidence intervals that correspond to 68.27%, 90% and 99% in one dimension.

(l) Also calculate the same confidence intervals in the case of two and three dimensions.